

AEROFATIGUE OPERATIONAL SAFETY

WORK INTENSITY MANUAL & FATIGUE MATRIX

Document Ref: AFM-WI-2026-V1 | Aviation Operational Safety System | Naval & Flight Ops

1. PURPOSE & SCOPE

This manual defines operational workload levels, fatigue risk factors, and mandatory mitigation procedures.

2. WORK INTENSITY CATEGORIZATION & SCORING CRITERIA

[+] LOW INTENSITY - Score Factor: 0 Points

- Scope: Routine ground handling, scheduled daily pre-flight walkarounds, light administrative reviews.
- 3. In-flight relief crew rotation is mandatory when mission flight time exceeds 7.5 consecutive hours.

[+] MEDIUM INTENSITY - Score Factor: +5 Points

- Scope: High turnaround operations, line defect rectification, intensive pre-flight avionics checks.

[+] HIGH INTENSITY - Score Factor: +10 Points

- Scope: Complex engine/gearbox changes, SAR emergency deployment, deck landing in rough seas.

3. CARTESIAN RISK BANDS & SUPERVISORY PROTOCOLS

LOW RISK (0 - 24 pts): Personnel fully fit for flight & maintenance tasks. Standard shift rotation.

MODERATE RISK (25 - 49 pts): Monitor duty cycle. Mandatory 15-min hydration & rest break after 4 hrs.

HIGH RISK (50 - 74 pts): Supervisor approval required. Re-evaluate consecutive duty days & rest.

CRITICAL RISK (75 - 100 pts): MANDATORY GROUNDING / STAND DOWN. Minimum 10h uninterrupted rest.

Formula: Score = Sleep Penalty (0-30) + Duty Duration (0-25) + Shift Circadian (0/20) + Consec Days (0-10) + Workload (0-10)

4. MANDATORY OPERATIONAL SAFETY DIRECTIVES

- Duty periods exceeding 12 hours must incorporate 1.5 hours cumulative recovery breaks.